

AI-Driven Social Engineering Emerging Threats to West African Organisations

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The Purpose Protocol — Cohort 1

1. Executive Summary

AI-driven social engineering uses generative AI to make scams more convincing, scalable, and personalised. Attackers use AI to write flawless phishing emails, clone voices, generate deepfake videos, and hold long conversations that build trust before asking for money or data.

This matters now because tools that once required technical skill and time are available for free or cheap online. For Nigerian and West African organisations, this increases risk to banks, universities, government offices, and SMEs where staff may not be trained to spot AI-generated content. A single attack can lead to financial loss, data breaches, and reputational damage. Traditional security controls focused on typos and bad grammar no longer work.

How AI Changes Social Engineering:

Traditional Method	With AI
Generic phishing emails with typos	Flawless emails written in your boss's tone, referencing real projects
Manual voice impersonation	AI voice cloning that sounds exactly like a CEO or family member
Basic fake profiles	Deepfake video calls + AI-generated LinkedIn profiles
Slow, 1-by-1 targeting	Automated scraping + personalized attacks on 1000s of people at once

The key shift: speed + personalization + realism.

2. Attack Overview

A step by step overview of how the attack work.

- 1. Data collection:** Attackers scrape LinkedIn, company sites, and social media for names, roles, projects, and writing style.
- 2. Content generation:** AI writes spear-phishing emails in the tone of a CEO, creates voice clones from 30 seconds of audio, or generates fake profiles to build rapport on WhatsApp and LinkedIn.
- 3. Delivery and engagement:** The attack is delivered via email, SMS, WhatsApp, or phone call. AI chatbots keep the conversation going to lower suspicion.
- 4. Exploitation:** The victim is tricked into transferring money, sharing credentials, or installing malware.
- 5. Cover-up:** Attackers use burner numbers and crypto wallets to make tracing harder.

Role of AI: AI removes language barriers, eliminates typos, and allows one person to run thousands of personalised attacks. It enables attacks that weren't possible at scale before, like realistic voice calls and context-aware phishing.

Real-world example: In 2023, a Hong Kong employee transferred \$25M HKD after a video call with what appeared to be the CFO and colleagues. The entire call was AI-generated. Source: BBC News

3. MITRE ATT&CK Mapping

- Tactic: Initial Access | Technique: T1566.001 - Phishing: Spear phishing Voice
Attackers use AI voice calls to establish first contact and build trust, bypassing email filters.
- Tactic: Initial Access | Technique: T1566.002 - Phishing: Spear phishing Link
AI writes personalised emails with malicious links that match the victim's context.
- Tactic: Credential Access | Technique: T1539 - Steal Web Session Cookie
After gaining trust, attackers direct victims to fake login pages to steal session tokens.
- Tactic: Impact | Technique: T1531 - Account Access Removal
Attackers lock victims out of accounts after stealing credentials to cover their tracks.

4. Who Is at Risk

Organisations with frequent external communication, financial transactions, and lower security maturity are most at risk.

High-risk sectors in West Africa:

- **Banks and fintechs:** Finance teams handle high-value transfers and are prime targets for impersonation.
- **Universities:** Students and staff are targeted with fake scholarship and job offers.
- **Government agencies:** Public officials have abundant online data for AI to use.
- **SMEs:** Limited IT staff and reliance on WhatsApp make verification harder.

Nigeria example: In 2025, a Lagos SME lost ₦12M after an employee responded to a WhatsApp message from a cloned "supplier" account. The AI chatbot maintained conversation for 2 weeks before requesting payment to a new account.

5. Detection Methods.

What to look for:

- Content anomalies: Emails are too perfect, lack internal jargon, or reference events you didn't attend.
- Behavioural red flags: Urgency, requests to bypass process, refusal to verify on a known channel.
- Technical signals: Lookalike domains, mismatched reply-to addresses, links to URL shorteners.
- Audio/video issues: Robotic speech, lip-sync lag, unnatural blinking in video calls.

Practical checks for security teams:

- Enable DMARC, SPF, DKIM to block email spoofing.
- Monitor for new domains similar to your company name.
- Review logs for logins from unusual locations after password resets.
- Use AI detection tools to flag synthetic media, but don't rely on them alone.

6. Prevention & Recommendations.

Technical controls:

- Enforce MFA on all accounts, preferably with passkeys or hardware tokens.
- Deploy email authentication and block external emails that fail DMARC.

People controls:

- Run quarterly training using real AI-generated phishing examples. Make it practical, not theoretical.
- Establish a call back policy for any financial or data request. Never verify via the same channel.

Low-budget recommendation for Nigerian organisations:

Create a shared "verification phrase" for finance and leadership teams. Change it monthly and never send it digitally. This stops 80% of impersonation attacks without cost.

7. References

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